A3-PS003-TC186

Federated Traffic-Flow Prediction with Communication-Cost Evaluation

PS-003 · Category: Smart Cities

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Problem Description

- **Problem.** Centralised traffic forecasting makes every intersection or roadside sensor stream raw readings to one server, using cellular bandwidth and concentrating mobility records.
- **Idea.** Federated Learning (FL) trains where the data lives and exchanges only model weights. But every round costs bytes, and traffic differs between intersections (*non-IID*), which can slow or degrade convergence.
- **Question.** Can K simulated intersections jointly learn a compact flow forecaster with FedAvg, and how do accuracy (MAE, RMSE) and *cumulative bytes* compare with centralised and local-only training, under IID and non-IID data?
- **Our solution.** A PyTorch simulator on one public traffic-*flow* dataset (PEMS04) split across $K{=}4$ virtual intersections, with a byte meter on every transfer, leakage-free chronological evaluation and 3-seed statistics.

Research questions

RQ1 FedAvg vs. centralised vs. local-only accuracy.
RQ2 Effect of non-IID data (volume, temporal pattern).
RQ3 MAE vs. cumulative bytes: what does accuracy cost?
RQ4 Do more local epochs / FP16 cut bytes to convergence?
RQ5 When does FL transfer more bytes than raw upload?

Scope and report status

Out of scope: live cameras, city-wide deployment, formal privacy guarantees (no DP / secure aggregation).
Status: this is a design report. *No training results exist yet.* Numbers shown are design parameters or analytical arithmetic and are tagged **[DESIGN ARITHMETIC]**; planned work is tagged **[PLANNED]**; measured results will appear as **[MEASURED]** only after runs complete.

Background

- **Traffic-flow forecasting** predicts near-future vehicle counts from recent history; the standard pipeline pools all sensor data centrally.
- **Federated Averaging (FedAvg)** [1]: in each round the server sends the global weights w^t; each client runs E local epochs of SGD on its own n_k training windows; the server returns a weighted average.
- **Communication cost** per round is $K \times$ (upload + download) model bytes; total cost = rounds $\times$ per-round cost. More local epochs can reduce rounds, but under non-IID data local models drift apart [3],[4].
- **Non-IID traffic** arises naturally: busy vs. quiet sites (volume), commuter vs. commercial peaks (temporal pattern).

FedAvg update (round t)

$$w_k^{t+1} = \mathrm{LocalSGD}_E\left(w^t;\, \mathcal{D}_k^{\text{train}}\right)$$

$$w^{t+1} = \sum_{k=1}^{K} \frac{n_k}{n}\, w_k^{t+1}, \quad n = \sum_k n_k$$

Raw data local $\neq$ private

Keeping raw records on the client is not a formal privacy guarantee: shared gradients and updates can leak training data in some settings [5]. This work adds no differential privacy or secure aggregation and **makes no privacy claim**; it studies accuracy and communication only.

Literature Review

Only the works that shape our design are listed. No IETF RFC is needed: we meter model-payload bytes, not protocol headers.

	Work	Relevant contribution	How we use it
[1]	McMahan et al., 2017	Introduces FedAvg (local SGD, weighted averaging)	Core algorithm; cited by the problem statement
[2]	Liu et al., 2020 (FedGRU)	FL-based GRU for traffic flow, with clustering	FL is viable for traffic flow; optional GRU ablation
[3]	Hsieh et al., 2020	Skewed partitions cause accuracy loss; skew degree drives difficulty	Justifies graded, quantified heterogeneity
[4]	Li et al., 2020 (FedProx)	Handles statistical heterogeneity with a proximal term	Future work only; we keep plain FedAvg
[5]	Zhu et al., 2019	Gradient-inversion leakage of training data	Basis for our no-privacy-guarantee statement
[6],[7]	Chen et al., 2001; Guo et al., 2019	PeMS source; PEMS04/08 benchmark channels	Data provenance and channel check

Positioning

We do not propose a new algorithm. Our contribution is a **controlled evaluation protocol**: same model, same data, chronological leakage-free splits, byte-exact metering in both directions, quantified non-IID levels, convergence-to-threshold in rounds *and* bytes, per-client errors, and a stated break-even assumption, all repeated over 3 seeds.

Proposed Solution: Simulator and Core Algorithm

Pipeline: verify data → chronological split → partition into $K{=}4$ clients → train (centralised, local-only, FedAvg) → meter bytes → evaluate → dashboard.

```
for t in 1..R:                          # rounds
  dn = encode(W_t)                      # FP32 or FP16
  for k in 1..K:                        # clients in turn
    meter.down[t] += len(dn)
    w = decode(dn)                      # received copy
    for e in 1..E:
      for batch in shard_k: w = SGD(w, batch)
    up_k = encode(w) + pack_i64(n_k) # + 8-byte n_k
    meter.up[t] += len(up_k)
  W_{t+1} = sum_k (n_k/n)*decode(up_k.w)
  log(t, val_MAE(W_{t+1}), meter.up[t],
         meter.down[t], meter.cumulative)
```

Serialization: flat parameter vector as little-endian bytes (`tobytes()`); order and shapes agreed in advance, no headers. FP16 is applied to both directions; aggregation stays FP32.

Design parameters [PLANNED]

Dataset	PEMS04, flow channel only
Clients	$K = 4$ (optional $K = 3, 5$)
Input	12 past flow values (60 min) + sin/-cos time-of-day of target
Target	flow 3 steps after last input (15 min ahead)
Model	MLP $14 \to 64 \to 32 \to 1$, ReLU: **3,073 parameters** (12,292 B FP32)
Training	mini-batch SGD, batch 512
Rounds	$R = 50$, all clients each round
Seeds	0, 1, 2
Stack	Python, PyTorch, Streamlit, Plotly

A small GRU is an optional ablation only.

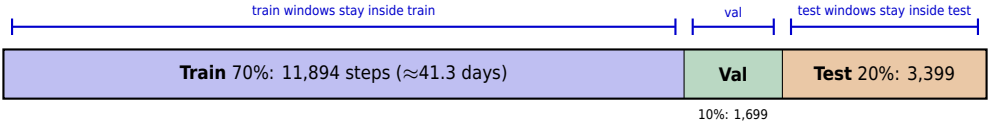
Proposed Solution: Dataset and Partitioning

Check	PEMS04 finding
Flow channel	Documented channels: flow, occupancy, speed per detector, 5-min aggregation [7]; 307 stations, 16,992 steps (Jan–Feb 2018). We use **flow only**; speed-only benchmarks (METR-LA, PEMS-BAY) were rejected. Verify array shape $16992 \times 307 \times 3$ on the downloaded file.
Source	Caltrans PeMS [6], free registration. The preprocessed benchmark file circulates via research repositories (e.g. `github.com/Davidham3/ASTGCN-2019-mxnet`, deprecated).
Terms: unverified	Not confirmed by us: we could not read the PeMS download terms. A third-party repository quotes them as public domain unless indicated, and Caltrans' general Conditions of Use govern its sites. The benchmark copy has no licence file. **Safest path:** download from PeMS under your own account (or rebuild District-4 5-min flow), keep data and derived arrays out of the repo, cite Caltrans PeMS, and ask the mentor to confirm the dataset is acceptable.
Limitation	Freeway detectors, not signalised intersections; a "virtual intersection" is a station group.
Data quality	Missing/constant sensors handled by one rule fixed before training, using no future data.

Partition	Rule (statistics from the training period only)
IID	Sensors shuffled by run seed, dealt evenly (77/77/77/76).
Non-IID-V (primary)	Sensors ranked by training mean flow, cut into 4 quartile groups (volume skew).
Non-IID-P (secondary)	k-means (k=4) on unit-sum daily profiles (peak shape/timing); unequal sizes, so n_k/n differ.

Heterogeneity [PLANNED]: H_W = mean W_1(client, pooled)$/\sigma_{\text{pooled}}$; H_P = mean cosine distance of daily profiles; H_V = max/min client mean flow; reported per partition.

Proposed Solution: Leakage-Free Evaluation Protocol



Leakage controls

- Chronological split. A window spans 15 steps (12 inputs, target 3 steps after the last input) and never crosses a boundary: 11,880 / 1,685 / 3,385 windows per sensor. Test inputs never use validation data.
- Z-score, volume ranking, *k*-means: training data only (a few scalars per client are shared once; not byte-metered).
- Tuning: learning rate $\in \{0.01, 0.03, 0.1\}$ on a separate tuning seed (100), chosen by pooled validation MAE, same grid for every method.
- Reported model = round/epoch with best validation MAE. Test curves are logged for plots but never used for selection.

Metrics and baselines [PLANNED]

- **MAE, RMSE** in vehicles per 5 min, pooled and *per client*; mean $\pm$ sample std over seeds 0, 1, 2; worst-client MAE.
- **Budgets:** centralised and local-only train 250 epochs, equal to FedAvg's $R \times E$ at $E{=}5$. Persistence and time-of-day average are reference models.
- **Convergence:** $\tau = 1.10\times$ centralised best validation MAE (same seed, partition), a rule fixed before any federated run. Report first round with pooled validation MAE $\leq \tau$ and bytes then; if never reached in R rounds, report *not reached*.
- **Central figure:** test MAE vs. cumulative bytes.

Optimization Proposed

Before: $E{=}1$, FP32. After: more local epochs or FP16. Fixed-round costs are analytical ($K{=}4$); convergence columns are measured later.

			Fixed-round cost (analytical)		To reach τ (measured)	
Configuration	E	**Prec.**	**B / round**	**B at $R{=}50$**	**Rounds**	**Bytes**
Reference FedAvg (before)	1	FP32	98,368	4,918,400	TBD	TBD
More local epochs (after)	5	FP32	98,368	4,918,400	TBD	TBD
More local epochs (after)	10	FP32	98,368	4,918,400	TBD	TBD
FP16 payload (optional)	5	FP16	49,200	2,460,000	TBD	TBD
Local fine-tune (optional)	–	–	+0	+0	–	–

TBD: not yet run (may be "not reached"); bytes to τ = rounds to $\tau \times$ B/round. FP32 per round: up $4(12{,}292{+}8) = 49{,}200$ + down $4 \cdot 12{,}292 = 49{,}168$ B; FP16: $4(6{,}146{+}8) + 4 \cdot 6{,}146 = 49{,}200$ B. Fine-tuning is reported separately.

Break-even vs. raw upload (analytical)

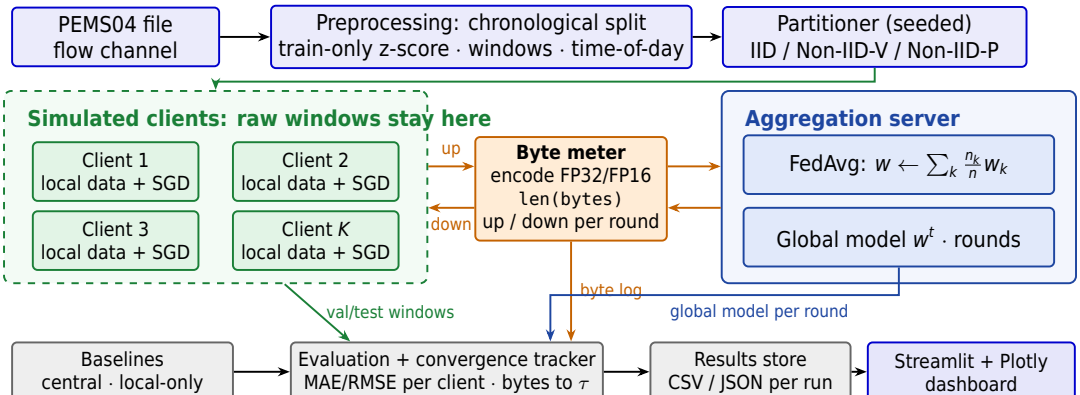
Raw: clients upload training flow once, $307 \times 11{,}894 = 3{,}651{,}458$ values = **14,605,832 B** (FP32) or **7,302,916 B** (uint16); validation adds about 14%. Break-even rounds (FP32 / uint16 raw): FP32 payload 148.5 / 74.2; FP16 payload 296.9 / 148.4. **Rule:** FL is cheaper only if *measured* bytes to τ are below the raw figure.

Caveats

Beyond break-even FL sends *more* bytes. PEMS04 is tiny aggregated counts, favouring centralised; video would shift this. Raw baseline omits timestamps, IDs, ongoing streaming.

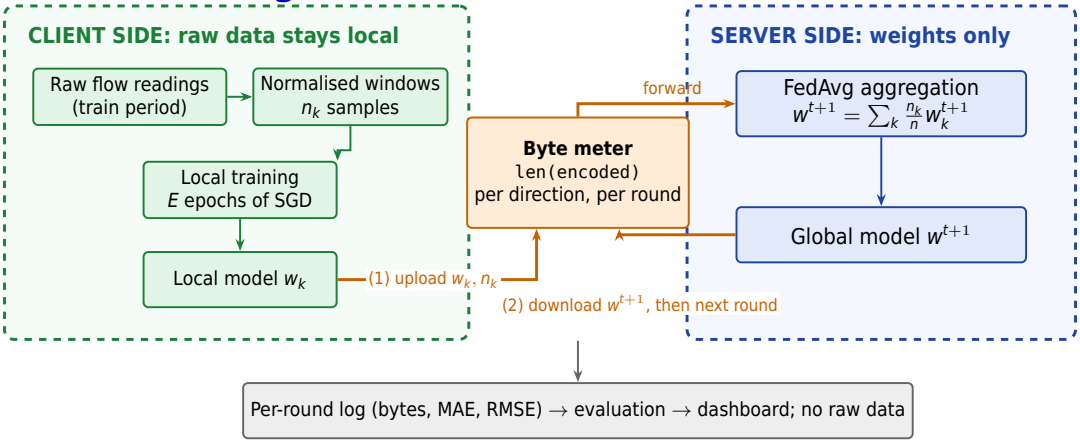
Experiments [PLANNED]: (A) 3 partitions $\times$ {centralised, local-only, FedAvg $E{=}5$}; (B) $E \in \{1, 5, 10\}$ on IID and Non-IID-V; (C) FP16, fine-tune if time permits; all $\times$ 3 seeds.

Solution Architecture & Design



Green: local raw data **Orange**: exchanged, metered model bytes **Blue**: server Single-process simulation.

Data Flow Diagram



Crosses the boundary: model weights (3,073 floats = 12,292 B in FP32) plus an 8-byte sample count, each direction, each round. **Does not cross:** raw readings, windows, labels.

No formal privacy guarantee. Weight updates can still reveal information about training data [5]; this design reduces raw-data movement but adds no DP, secure aggregation or encryption.

Deployment Timeline and Success Measures

Project steps in order. “Deployment” = runnable simulator and dashboard, not a field deployment.

Step	Milestone	Work items	Exit check
1	Repo, data	Private repo, add `aiori-hackathon`; data check	Shape checked; terms pending
2	Pipeline	Split, scaling, windows, partitioners	Tests: no window crosses a split
3	Baselines	Centralised, local-only, persistence	Validation table
4	FedAvg, meter	FedAvg loop, byte meter, tracker	Logged 98,368 B/round (FP32)
5	Experiments	Matrices A, B (3 seeds); optional C	All runs in results store
6	Dashboard	Streamlit + Plotly, stored results only	MAE-vs-bytes plot renders
7	Report, demo	Final report; mentor demo	PDF in repo

Success is measured by [PLANNED]

- MAE/RMSE tables, pooled and per client, 3 seeds
- MAE-vs-cumulative-bytes curves per partition
- Rounds and bytes to τ, or explicit “not reached”
- Expected IID/non-IID gaps are *untested*

Risks and limits

- Freeway data, one region, two months
- Single-process simulation; headers excluded
- Dataset terms unconfirmed; data not redistributed
- No formal privacy guarantee

Bibliography

[1] B. McMahan, E. Moore, D. Ramage, S. Hampson, B. Agüera y Arcas. “Communication-Efficient Learning of Deep Networks from Decentralized Data.” *AISTATS 2017*, PMLR 54:1273–1282. https://proceedings.mlr.press/v54/mcmahan17a.html

[2] Y. Liu, J. J. Q. Yu, J. Kang, D. Niyato, S. Zhang. “Privacy-Preserving Traffic Flow Prediction: A Federated Learning Approach.” *IEEE Internet of Things Journal* 7(8):7751–7763, 2020. doi:10.1109/JIOT.2020.2991401

[3] K. Hsieh, A. Phanishayee, O. Mutlu, P. Gibbons. “The Non-IID Data Quagmire of Decentralized Machine Learning.” *ICML 2020*, PMLR 119:4387–4398. https://proceedings.mlr.press/v119/hsieh20a.html

[4] T. Li, A. K. Sahu, M. Zaheer, M. Sanjabi, A. Talwalkar, V. Smith. “Federated Optimization in Heterogeneous Networks.” *MLSys 2020*. arXiv:1812.06127

[5] L. Zhu, Z. Liu, S. Han. “Deep Leakage from Gradients.” *NeurIPS 2019*, pp. 14774–14784.

[6] C. Chen, K. Petty, A. Skabardonis, P. Varaiya, Z. Jia. “Freeway Performance Measurement System: Mining Loop Detector Data.” *Transportation Research Record* 1748:96–102, 2001. doi:10.3141/1748-12. Data portal: https://pems.dot.ca.gov

[7] S. Guo, Y. Lin, N. Feng, C. Song, H. Wan. “Attention Based Spatial-Temporal Graph Convolutional Networks for Traffic Flow Forecasting.” *AAAI 2019* (33rd). Dataset description: https://github.com/Davidham3/ASTGCN-2019-mxnet

Thank You for Your Attention!

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